Test 8 – Grade 8

Name: _____
Grade / Class: 8

Section A: English Language & Literature

1. Reading Comprehension
Read the passage and answer the questions:

"During the winter festival, Omar volunteered to organize a recycling campaign at his school. He taught students to separate plastics, paper, and metals. His team also created posters highlighting the importance of reducing waste. By the end of the week, the school had collected a significant amount of recyclable materials, and Omar felt satisfied with his efforts."

a) What activity did Omar organize during the festival?
b) Name two tasks Omar performed while volunteering.
c) How did Omar feel at the end of the campaign?
d) Summarize the passage in four sentences using your own words.

2. Grammar & Vocabulary
a) Combine these sentences using a suitable conjunction:
"The sun set behind the mountains. The hikers decided to set up camp."

b) Identify all nouns in the sentence:
"The talented musicians performed beautifully during the annual cultural festival."

c) Correct the error:
"Each of the students have submitted their projects on time."

d) Change into reported speech:
"She said, 'I will read all the books in the library this week.'"

3. Writing
Write a short essay (120–150 words) on the topic: "The Importance of Environmental Conservation". Include benefits, challenges, and personal opinion.

Section B: Mathematics

1. Algebra
a) Solve for x:
$7x - 10 = 3x + 18$

b) Factorize:
$x^2 - 11x + 24$

c) Simplify:
$5m^2n - 7mn^2 + 2m^2n$

2. Geometry & Mensuration
a) A parallelogram has base 20 cm and height 15 cm. Find the area.
b) A sphere has a radius of 6 cm. Find its volume.

3. Ratio, Proportion & Percentage
a) A laptop originally costs $500. It is sold at a 12% discount. Find the selling price.
b) If 15 pencils cost $45, find the cost of 40 pencils.

4. Data Handling
The scores of 10 students in a science test: 12, 14, 15, 16, 13, 15, 14, 16, 17, 15.
a) Find the mean, median, and mode.
b) Represent the data in a simple frequency table.

Section C: Science

1. Biology
a) What are the main functions of the human liver?
b) Explain the process of transpiration in plants.
c) List two differences between reptiles and amphibians.

2. Physics
a) A train accelerates from 10 m/s to 50 m/s in 20 seconds. Calculate acceleration.
b) Define potential energy and provide an example.
c) Explain the difference between reflection and diffraction of light.

3. Chemistry
a) Identify whether the following are element, compound, or mixture:

- **Iron**
- **Carbon monoxide**
- **Sea water**

b) Write a balanced chemical equation for the reaction between zinc and sulfuric acid.

4. Environmental Science
a) List two water pollutants and their effects.
b) Suggest three practical methods to conserve energy at home.

Section D: Social Studies

1. History
a) Name two major revolutions of the 20th century and their outcomes.
b) Explain the significance of the Industrial Revolution.
c) Identify the Allied Powers in World War I.

2. Geography
a) Define longitude and explain how it affects time zones.
b) Name the largest rainforest in the world and its location.
c) Discuss two environmental consequences of mining activities.

3. Civics / Political Science
a) Differentiate between parliamentary and presidential systems of government.
b) State two duties of citizens in a democratic society.

4. Economics
a) Explain the concept of scarcity with an example.
b) Differentiate between revenue and profit in business.

Section E: ICT / Computer Studies

1. Computer Basics
a) Define hardware and give one example.
b) Name one input device and one storage device.

2. Applications
a) Which spreadsheet function calculates the average of a dataset?
b) Mention two uses of database software in schools.

3. Programming Basics
a) Explain conditional statements in programming with an example.
b) Write a Python code snippet to print all even numbers from 2 to 30.

4. Cyber Awareness
a) What is phishing and how can it be avoided?
b) List two ways to create strong passwords.

Section F: Islamiyat

1. Quran & Hadith
a) Name two moral teachings in the Quran.
b) Narrate a Hadith that encourages forgiveness.

2. Aqidah / Belief
a) List the six articles of faith in Islam.
b) Explain why belief in the Day of Judgment is important.

3. Fiqh / Worship
a) Why is Salah essential for a Muslim?
b) Describe the benefits of Sadaqah (charity).

4. Islamic Ethics & Morality
a) How does Islam promote honesty in business?
b) Explain the importance of cleanliness according to Islam.

Section G: Arts & Music

1. Visual Arts
a) Explain how shading is used in drawing.
b) Name two tertiary colors.

2. Music
a) Define rhythm and tempo in music.
b) Give an example of a percussion instrument.

Section H: Health & Physical Education

1. Physical Fitness
a) Name two exercises to strengthen the core muscles.
b) List one team sport and one individual sport.

2. Health & Hygiene
a) Why is proper sleep important for teenagers?
b) Mention two ways to maintain personal hygiene.

3. Safety & Environment
a) State one safety rule when handling chemicals in a laboratory.